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Review

Effect of Waterlogging on Growth and Productivity of Fruit Crops

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Abstract: The production of fresh fruit as well as olive orchards is increasing around the world, in order to meet the global demand for both fruits and olive products. This results in the spread and establishment of fruit and olive tree cultivation in areas where they were not found before, for example, plains and lowlands prone to waterlogging. Climate change is having a significant impact on the natural environment. Agricultural open-field crops have less growth and yield under these harsh weather conditions. Nowadays, unpredictable rainfall more often exposes field crops to waterlogging on a regular basis. This is a very stressful factor which can cause a reduction in yield and even total crop elimination. In this review, the morphological and physiological parameters affected by waterlogging are developed in order to understand better how olive and other fruit crops respond to waterlogging conditions and how this affects their development and productivity. Having a better understanding of these mechanisms can help us design strategies and approaches to increase fruit crop resistance to waterlogging stress.

Keywords: aerenchyma; climate crisis; flooding; hypoxia; rainfall; respiration; stress

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1. Introduction

The Mediterranean ecosystem is characterized by high temperatures, sunlight levels, and vapor deficits, while it lacks precipitation during summer months [1]. On the other hand, the Intergovernmental Panel on Climate Change (IPCC) has predicted increased extreme weather events for much of the world's crop-producing arable region [2]. This indicates that the climate is changing, resulting in higher temperatures and changes to the water cycle, as shown by erratic and heavy rainfalls that can cause waterlogging and flooding [3]. Extreme weather events present a serious challenge for agricultural food crops in these times of climate change, resulting in considerable yield losses and, in some cases, crop annihilation [4]. The rising temperatures create warmer air masses that hold more water, which enhances the increasing frequency and magnitude of local rainfall [5]. Flash floods and waterlogging have terrible ecological and social consequences and they are frequently caused by torrential rain [6].

Waterlogging is globally one of the main abiotic stressors that can significantly reduce yields and has become a serious concern [7]. This is a problem that we mostly face in field crops; however, it can also happen in perennial crops, such as fruit and olive trees. Flooding (water remaining above the soil surface) and waterlogging (the soil being water-saturated) lead to oxygen deficiency in the soil because water entering the soil removes oxygen-rich air from soil pores [8]. Incidents of flooding in fruit trees have increased as a result of climate change [9]. According to Insausti and Gorjón [10] experimenting with waterlogged peach trees, during the lignification stage of the fruit endocarp and the beginning of its exponential growth period, fruit diameter growth was low, resulting in reduced fruit production. Moreover, flooding causes several problems in the soil environment, as the air present in the soil is replaced by water [11]. More specifically, the limitation of oxygen in roots causes hypoxic conditions, causing the change of their metabolism from aerobic to anaerobic [12]. Under these circumstances, acclimatization to the amount of stress itself

and the process of recovering from a stressful scenario both affect a plant's resilience to bad environmental conditions. Plants must continually control their metabolism in order to keep on growing and developing in a very dynamic and unfavorable environment [13]. On the other hand, the main measure that should be implemented to avoid waterlogging is drainage, which is obviously the most important technique to fight waterlogging: a careful design of the draining system is essential [14].

2. What Is Waterlogging and What Are Its Impacts on Plant Physiology?

Waterlogging is one of the major abiotic stresses that limits crop productivity and the growth of many crops all over the world [15]. In essence, waterlogging is the phenomenon in which free water overlays the soil surface of cropland, causing unfavorable conditions for plants to grow in, either directly by affecting root metabolism or indirectly through decreasing the availability of vital nutrients to the plant [16].

Essentially, waterlogging, soil erosion, water flow, and pedoturbation all have an impact on the physical and chemical properties of soil, as well as on its biochemical features (Figure 1). The soil hydroperiod lowers soil oxygen levels and hence prevents general oxidative reactions, which are essential to the breakdown of organic matter [17]. Furthermore, soils that have been exposed to excessive water show changes in their pH, electrical conductivity, and oxidation-reduction potential (ORP) [18]. Additionally, flooded soils significantly restrict machinery operations, impacting soil preparation, plant care, and crop harvesting, resulting in reduced yields [19].

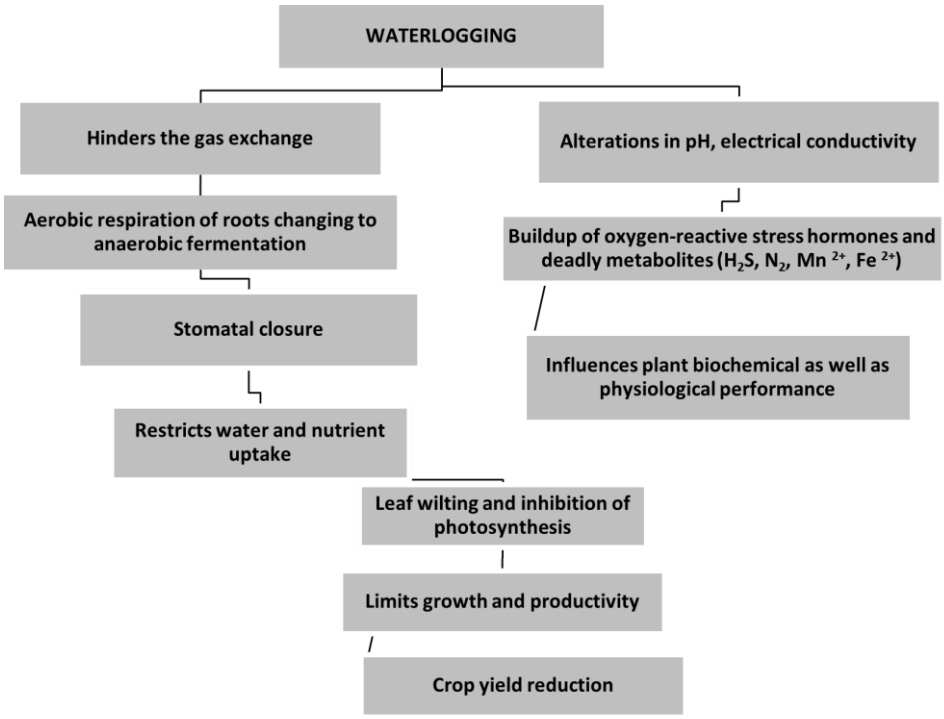

Figure 1. Stress caused by waterlogging, its effects on soil and plants, and how different plants respond to these challenging circumstances in terms of metabolism [4].

In the case of direct effects, water occurrence hinders the gas exchange between plant roots and the atmosphere [20]. As is known, plants are aerobic organisms that require O_2 for the absorption of nutrients [21]. In waterlogged soils, there is an intense lack of oxygen in the roots of plants due to its replacement with water (Figure 2). More specifically, pores saturated with water rapidly deplete oxygen levels, with the remaining oxygen swiftly consumed by roots and microbes. Clay soils, characterized by their fine pores, are particularly vulnerable to oxygen deficiency compared to sandy soils, which maintain aeration even when submerged in flowing water [22]. Waterlogging causes a significant

drop in soil oxygen levels, as the rate of oxygen diffusion in waterlogged soils is 10,000 times slower than in well-drained soils [23]. This condition leads roots from aerobic respiration to anaerobic fermentation.

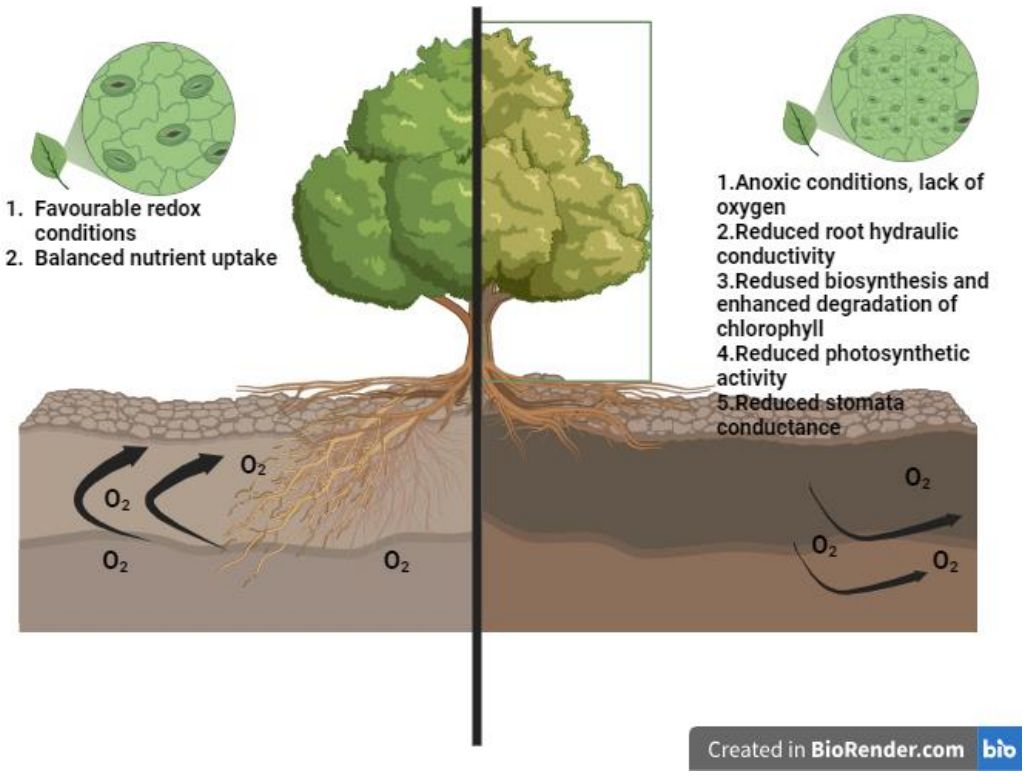

Figure 2. A plant's waterlogging physiology. An unwaterlogged tree in well-oxygenated soil can be seen on the left. A tree under waterlogging stress in an anoxic soil is shown on the right, along with a list of typical physiological reactions to waterlogging (BioRender.com; accessed on 25 April 2023).

In coping with oxygen deprivation, plants have evolved various morphological, anatomical, and metabolic adaptations to safeguard against cell damage in such unfavorable environments [24]. Waterlogging impacts stem growth based on the plants' adaptability levels. Some plants can either mitigate or exacerbate the effects even by positioning their flowers and leaves above the water surface [25,26]. Nonetheless, for plants vulnerable to waterlogging like papaya (*Carica papaya*) [27], just 48 h of waterlogging can significantly hinder both stem and root growth in terms of diameter and length [28]. Also, according to Kozlowski and Pallardy [29], waterlogging reduces root and stem growth of woody plants, and moreover, it changes nutrient uptake and carbohydrate translocation, thus enhancing senescence and plant mortality.

Waterlogging affects leaves similarly to water stress, resulting in lower stomatal conductance and elongation rate [30]. The leaves turn yellow and develop more necrosis as a result of prolonged exposure to hypoxic stress on the roots, which is followed by their abscission. The roots' inability to absorb and transfer nutrients and water is what causes these symptoms [29,31,32].

The negative consequences of long-term waterlogging can be observed during all growth stages of a plant throughout its lifecycle, ultimately resulting in a reduction in its productivity [33–37]. Crane et al. (2020) [27] point out that waterlogged trees manifest different symptoms, diminishing both growth and yield, and impacting not only non-tolerant but also tolerant fruit species. Symptoms of waterlogging on fruit trees progress gradually, as outlined below: (1) wilting and scorching leaves; (2) fruit drop, leaf chlorosis and drop; (3) regressive death of the stem and limb dieback; (4) death of the tree [27]. Although it is difficult to give a precise time course that applies to all fruit crops for the series of events that occur during flooding, the responses of these crops can be divided into

early, middle, and late responses. Whereas the sequence of flood responses is similar for all fruit crops, the specific time required to elicit each response varies for each species based on inherent flood tolerance [38].

The root's anaerobic stress limits its respiration, leading to reduced energy recovery in the process [29], an upsurge in fermentation, cytoplasmic acidosis, or acidity of the cytosol, and toxic consequences [39]. This energetic deterioration affects many metabolic processes, such as root cell ATP synthesis [40]. More specifically, during fermentation, the production of adenosine triphosphate (ATP) amounts to two ATP molecules per glucose molecule through substrate phosphorylation [41]. In contrast, oxidative phosphorylation yields thirty-two ATP molecules [42], resulting in energy production that is 16 times less efficient when using fermentation.

In the case of indirect effects, waterlogging decreases Na^+ concentration while it increases K^+ content in both leaf and root tissues, which causes an energy shortage and therefore reduced ion uptake and root growth [43]. Faulty root respiration inhibits root growth and transporter-driven ion uptake, resulting in an energy shortage and reduced ion uptake, leading to less energy for root growth. Furthermore, in basic-reactive soils, exposure to brief hypoxia in the roots lowers the redox potential of the soil and increases the availability of micronutrients by converting Fe^{3+} to Fe^{2+}, which is uptaken by plants. According to Schaffer et al. [44], this promotes carambola (*Averrhoa carambola*) flowering and fruit production.

Stomatal closure in response to hypoxic stress from the root system has been observed in various species, such as lemon [45] and strawberry [46], as a mechanism to reduce water loss [41]. Moreover, flood-sensitive species seem to lack a mechanism to reopen stomata that closed due to hypoxic soil conditions [47]. Intriguingly, waterlogging can not only elevate stomatal resistance but also limit water uptake by reducing root hydraulic conductivity [48,49], resulting in an internal water deficit in the plant [50,51] (Figure 2).

Photosynthesis is primarily affected by stomatal closure, a decrease in Rubisco enzyme activity, and chlorophyll destruction, leading to desiccation symptoms and growth delays [52]. As a result, the crops cultivated under this stress are significantly impacted and suffer from various unfavorable conditions. More specifically, as stomatal closure increases, the rate of photosynthesis decreases, and as leaf drop increases, the rate of photosynthesis declines again [47]. For instance, stomatal closure due to impaired root function during waterlogging will limit water and nutrient uptake and prevent carbon dioxide from entering the leaf, causing leaf wilting and senescence, inhibition of photosynthesis, and lower biomass production [33,53,54]. Several species, including lemons, have shown evidence of stomatal closure because of hypoxic stress on their root systems [45].

The decrease in the leaf chlorophyll index (LCI) (SPAD 501 m (Minolta, Inc., Kyoto, Japan) between plants in the flooded and nonflooded treatments suggests that this is an early reaction and could be a valuable, fast quantitative measure of flooding stress. Unquestionably, lower LCI was associated with lower net CO_2 assimilation (A) (LI 6400 XT, Licor Corporation, Lincoln, NE, USA) [55].

Waterlogging stress can result in physiological disorders such as impaired hormonal balance, degraded leaf capacity for photosynthesis, and harmed cells and membrane systems, as well as insufficient uptake of water, nutrients, and minerals, leading to poor plant development. The closure of stomata, which is related to plant photosynthetic efficiency, is also caused by waterlogging, disrupting gas exchange and ultimately diminishing yield and productivity [56]. Furthermore, plants exhibit a decline in stomatal conductance (g_s), often resulting from the reduced assimilation of net CO_2 and leaf chlorosis [57]. The limited uptake of water and various nutrients (such as P, Ca, Mg, Fe, Mn, Mo, etc.) reduces plant growth and development, and the accumulation of organic matter in the soil, in turn, leads to a decrease in net CO_2 accumulation [58,59].

What is more, waterlogging-induced stress has a negative impact on photosynthetic enzymes' activity, alters the chloroplast structure, and damages the photosynthetic reaction centers [60,61]. Water-soaked growing plants have shown a decrease in chlorophyll content,

particularly chlorophyll a and b, leading to a reduction in the photosynthetic rate and eventually in crop yield and production [62]. Furthermore, under waterlogging conditions, several reactive oxygen species (ROS) are generated due to excessive reduction in the electron transport chain, thus resulting in oxidative damage [63].

As mentioned above, waterlogging, flooding, or inundation affect the oxygen distribution in tissues and the exchange of various gases between cells, restricting oxygen exchange and respiration in mitochondria (aerobic respiration), and thereby influencing the typical biochemical and physiological performance of plants [64,65]. Lethal substances, such as aldehydes and alcohols, accumulate up in the tissues because of the decreased energy production [66]. Hormone metabolism is also impacted by anaerobic root conditions. Since root ethylene flows more slowly in wet soil than in well-aerated soil, root ethylene concentrations rise [30]. One known signaling molecule that aids in the acclimatization process of plants to low oxygen is ethylene [24]. Similar to this, the production of abscisic acid in flooded roots causes stomatal closure in leaves and is also involved in the development of aerenchyma in roots [67]. For instance, according to Perez-Jimenez and Perez-Tornero (2021) [68], citrus plants are negatively affected by hypoxia due to their inability to develop aerenchyma or hypertrophic lenticels.

Waterlogging can hinder fruit tree growth and reproductive development by inhibiting root respiration and accumulating harmful chemicals [67]. Inhibited processes include the creation of flower buds, flowering, and fruit set and growth [29]. For instance, compared to fruits from non-waterlogged trees, peach (*Prunus persica*) fruits were smaller after being waterlogged for 12 h a day for eight weeks, which causes ethylene to be produced and the pulp to soften early in the postharvest process [10]. Moreover, earlier studies on waterlogged apple cv. Stayman Winesap plants revealed that the fruits were small with somewhat high color, dropping early [69]. It was observed that flooded highbush blueberry plants had smaller fruits and reduced soluble solid content of the fruit compared with nonflooded plants ([38] and references therein).

Waterlogging is a problem that can impact olive trees, particularly those grown in areas with high rainfall or poorly drained soils. Olive trees are sensitive to waterlogging, and prolonged exposure to excess water can have significant negative effects on tree growth, yield, and fruit quality [70]. One of the primary effects of waterlogging on olive trees is reduced shoot growth and vigor [71]. When the soil is saturated with water, the oxygen supply to the roots is limited, which can lead to root decay and death. This can cause the tree to lose its ability to absorb nutrients and water from the soil, resulting in stunted growth, reduced canopy density, and lower yields [71]. Excess water can also cause fruit drop, reduce fruit size, and negatively impact the oil content and quality of olive fruits [45]. In severe cases, waterlogging can also increase the susceptibility of olive trees to pests and diseases [72].

Furthermore, wet soil, particularly where there is standing water, makes an excellent culture medium for numerous bacterial and fungal pathogens [73]. This strong combined stress weakens plants, especially their roots, which are then more vulnerable to disease, notably in the case of papaya, lychee (*Litchi chinensis*), pineapple (*Ananas comosus*), and avocado [74].

3. Morphological and Physiological Adaptations in Fruit Crops

There are several developmental responses to waterlogging that vary between roots and shoots and among different species. Crop species exhibit diverse reactions to waterlogging, with some responses regarded as adaptations (such as aerenchyma formation), while others are considered as injuries (e.g., chlorosis) [75]. Reduced growth is the outcome of O_2 deprivation because basic processes of development, physiology, and metabolism, including cell division, respiration, growth, nutrition and water intake, and transpiration, depend on O_2 [76].

Roots are among the most sensitive parts of a plant subjected to flooding [77]; root growth stops, or parts of them are damaged or die [22]. To maintain their function, roots respond to hypoxic conditions by changing in a variety of morphological and structural ways. One such adaptation is the development of aerenchyma in root tissue, which makes

it easier for plants to engage in aerobic respiration by enabling the transfer of gases from the above-ground shoot to the roots [65,78]. However, some species can produce new adventitious roots closer to the surface, adapting in this way to low oxygen levels by facilitating gas exchange [79]. Also, stem growth is impacted by waterlogging according to the plants' level of adaptability [9].

Waterlogging obstructs gases from escaping the roots through the soil, resulting in accumulation of ethylene in roots [80], decreased root growth [81], stimulation of auxin biosynthesis, and root gravitropism [82]. Due to the crucial role that ethylene and reactive oxygen species (ROS) play in the waterlogging response, various genes involved in ROS production have been recognized, including respiratory burst oxidase homolog (RBOH) expression [83] and ethylene response factors (ERFs) [75]. More specifically, many common molecules, including O_2, H_2O_2, and OH^-, belong to the class of ROS [84]. One of the primary ROS thought to be present, in multiple abiotic stresses such as hypoxic stress, repeated heat stress, and especially waterlogging, is H_2O_2, since it is comparatively stable in cell signaling [85]. Hydrogen peroxide performs a signaling role in plants by taking part in several signaling cascades, abiotic stress reactions, and programmed cell death [86]. For instance, H_2O_2 functions as a second messenger in plant responses to hypoxic stress and as a signaling molecule in response to repeated heat stress [87]. In order to keep the redox balance in plant cells, H_2O_2 can also change the expression of genes governing antioxidant enzymes and other transcriptional factors [88]. For instance, according to Christianson et al. (2010) [89], after five hours of hypoxia treatment, nearly 2000 differently expressed genes were found in flood-tolerant poplar trees when compared to roots under normal oxygen conditions. Furthermore, the waterlogging stress was found to have a strong transcriptional impact on several ABA pathway-related genes. For instance, the ABA degradation enzyme, ABA 8-hydroxylase, was increased while the gene encoding the rate-limiting enzyme in ABA production, *9-cis-epoxycarotenoid dioxygenase 1* (NCED1), was downregulated [90]. Waterlogging causes ABA to accumulate, and more ABA causes H_2O_2 accumulation, stomatal closure, the activation of genes for antioxidant enzymes, and modifications in plant actions of antioxidant enzymes [91]. The phytohormone ABA is known to react to cold, salinity, and drought conditions; waterlogging often leads ABA levels to decrease [90]. A reduction in ABA concentration would cause ethylene to build up, which would then encourage the growth of adventitious roots and aerenchyma in submerged plants [92,93]. These findings suggest that ABA has a negative regulatory effect on plants' ability to tolerate waterlogging stress [90].

Additionally, in waterlogged roots of *Cerasus sachalinensis*, elevated transcription of genes has been observed, linked to starch metabolism and the glycolysis/gluconeogenesis pathways [94]. According to Teoh et al. (2022) [90], one day of waterlogging greatly increased the expression of genes involved in glycolysis, such as *hexokinase*, *phosphoglycerate kinase*, and *pyruvate kinase*. It is not surprising that carbon metabolism was differentially affected under waterlogging stress given the necessity to retain enough energy for plant life since nutrient uptake is restricted in hypoxic roots. Interestingly, melatonin plays a role in controlling plants' reaction to waterlogging [95]. Moreover, melatonin biosynthesis genes of plants that endure waterlogging stress are elevated, causing the endogenous melatonin levels to increase [96]. For example, to preserve redox balance, exogenous melatonin treatment decreased H_2O_2 concentrations in the roots and leaves of peach seedlings and increased the activity of several types of antioxidant enzymes under waterlogging stress [97].

Another defense mechanism against hypoxia or anoxia is the increase in leaf nitrogen concentration [98] due to its ability to modulate traits that affect whole carbon fixation, such as saturated photosynthesis (Li 6400P photosynthesis system, Li-Cor, Lincoln, NE, USA) and leaf area [99].

According to Perez-Jimenez and Perez-Tornero (2021) [68] in an experiment contacted on citrus rootstocks, it was observed that one of the three genotypes, Cleopatra, exhibited a significant sensitivity to waterlogging stress, displaying alterations in the relative water content, stomatal conductance, and notably, photosynthesis (LI-6400, Li-Cor, Lincoln, NE,

USA). Several of these changes were reported even during the recovery phase, and it is worth investigating the extent of irreversibility of these parameters. On the contrary, the other two citrus rootstocks used did not exhibit such persistent effects, as they presented signs of recuperation in parameters that had decreased during waterlogging stress.

In another experiment applied to olive trees (*Olea europaea* L., cv. Arbequina), it was found that the combined stresses of salinity and waterlogging (hypoxia) had the greatest negative impact on olive tree growth and survival [71]. Additionally, it was observed that olive trees became less tolerant of salinity when subjected to waterlogged conditions. Specifically, 55% of the 341 monitored olive trees were dead 3.5 years after their plantation. The determination of threshold values can be useful in defining the tolerance of olive cv. Arbequina to salinity and waterlogging stresses [71].

The ability of roots to adapt and maintain their function in waterlogged conditions is crucial for the survival of plants, particularly in areas with high levels of precipitation or poor drainage. Understanding these adaptations can help inform the development of crop varieties that are better suited to waterlogged conditions and contribute to the development of more sustainable agricultural practices [13].

4. Solutions Reducing the Waterlogging Stress

Given the current context of rapidly changing and unpredictable environmental conditions, the resilience of plants is a critical factor for their survival. In addition to adapting to stress levels, plants also need to be able to recover from stressful conditions. This is especially important in environments that are dynamic and prone to fluctuations, where plants must constantly regulate their metabolism in order to maintain growth and development.

As the world population grows and the demand for food increases, there is a pressing need for researchers and plant breeders to develop more resilient crops and improve agricultural practices. This presents a major challenge for the future, requiring concerted efforts to produce crops that can tolerate a variety of environmental stresses, such as waterlogging, which can have a significant impact on crop growth and productivity. Despite the availability of data on numerous abiotic pressures, the impacts of waterlogging stress have received inadequate study.

Plants that can persist in waterlogged soils and maintain a high rate of photosynthesis often exhibit a range of anatomical and morphological changes. These changes include the production of aerenchyma in root tissues, the development of adventitious roots, and the formation of a barrier to root radial oxygen loss (ROL) [78]. Researchers have identified these adaptations as key factors in the ability of plants to tolerate waterlogging stress. Studies have shown that the development of aerenchyma, meaning air-filled spaces in plant tissues, allows the efficient diffusion of gases, such as oxygen, to areas that would otherwise be oxygen-deprived. Similarly, the emergence of adventitious roots provides additional surface area for gas exchange, while the barrier to ROL contributes to the prevention of oxygen loss from the roots [27,78,100].

To minimize hypoxic stress in fruit trees that require grafting, more tolerant rootstocks are selected [101]. Using rootstocks from other species, such as plum (*Prunus domestica*) and some peach and apricot (*Prunus armeniaca*) varieties, has led to greater resistance to this problem [102–104]. According to Kongsri et al. (2020) [105], guava plants produced by seedlings are more tolerant to waterlogging than those propagated by branch stacking. This indicates that the type of rootstock used is significant in adapting to waterlogging. Also, regarding their capacity to withstand flooding stress, the six Prunus rootstocks that were examined differed significantly from one another. The current standard rootstock for the subtropics, Flordaguard, does not seem to be as tolerant of flooding stress as the rootstocks MP-29, P-22, and R5064-5. Flordaguard's susceptibility to flooding stress indicates that subtropical regions that experience root zone flooding require different rootstocks. While P22 demonstrated high levels of ROS accumulation and R5064-5 has not yet been released for farmer trials, the rootstock MP-29 is presently available to growers [55].

Applying solid fertilizer, such as calcium oxide (CaO) or magnesium oxide (MgO), can enhance the redox potential of flooded soil. In addition to the soil's increased oxygenation, adding 5 g of magnesium peroxide (MgO) to the potting media before flooding papaya (*Carica papaya*) boosts the plants' total dry weight and leaf area when compared to nonflooded plants [106]. Redox potential, a proxy for O_2 content, is enhanced as a result [107].

In a similar vein, mycorrhizal colonization of roots can boost a plant's resistance to waterlogging by fostering biomass and growth through improved nutritional conditions and potential adaptation [108]. The study conducted by Chebet et al. (2020) [109] revealed that mycorrhizae (*Glomus caledonium*, *G. etunicatum*, *Gigaspora margarita*, and *Sceptellospora* sp.)-inoculated purple passionfruit (*Passiflora edulis* f. edulis) plants exhibited enhanced leaf retention, elevated proline and chlorophyll content in leaves, and prolonged maintenance of the foliar content of N and P. These authors hypothesized that mycorrhized plants' growing dry and fresh root weights enhanced their root health and promoted the uptake of mineral nutrients.

Applications of hydrogen peroxide (H_2O_2) or glycine betaine (GB) on the leaves of plants increase their capacity to withstand wet conditions, which facilitates their adaptation. Better responses in terms of growth and physiology (stomatal conductance, maximum photochemical efficiency of PSII [Fv/Fm], leaf water potential, relative water content, chlorophyll content, and net photosynthesis) are shown by the cape gooseberry (*Physalis peruviana*), which was flooded for 6 days with applications of 100 mM GB or H_2O_2 [110]. Generally speaking, measuring the impact of waterlogging stress can be performed quickly and easily using chlorophyll fluorescence [111–113].

Pruning branches after waterlogging can restore the balance of the shoot/root ratio in the tree, especially for injured roots. Pruning avocado trees can help them recover faster from hypoxic stress [49]. Additionally, the smaller leaf area restricts transpiration, which makes it challenging for plants with damaged roots to do so.

Concerning olive trees, in order to mitigate the impacts of waterlogging stress, it is important to take measures to improve soil drainage and aeration. This can be achieved by incorporating organic matter into the soil, installing drainage systems, and avoiding over-irrigation. Additionally, olive growers can establish their orchards in areas with well-drained soils or can select olive tree cultivars that are more tolerant to waterlogging such as Arbequina [71]. Other practical solutions could be planting olives in raised beds or applying mulch [71].

Despite the progress made in understanding the response of plants to waterlogging stress, there is still a significant gap between research in the laboratory and field conditions. While controlled conditions may simulate some aspects of waterlogging stress, field trials are essential to validate the findings in real-world scenarios. However, conducting field-based work is not without challenges, including soil leveling and access to water sources. Some studies have attempted to bridge the gap between controlled and field conditions, and they have shown that some plant physiological and morphological traits that are influenced the most by waterlogging were significantly correlated between the two settings [114,115]. To overcome these challenges, advances in genetic analysis and plant phenotyping are crucial for developing crops that are more resilient to waterlogging stress, which could ultimately lead to a more food-secure future.

5. Conclusions

Waterlogging is a serious issue that affects fruit and other crop yields and productivity across the world. Different plant species and genotypes respond to waterlogging stress in different ways throughout the long and short term. Some species can withstand waterlogging, whereas others are vulnerable to it. Under such circumstances, resistant species might flourish because they acquire specific adaptations that enable them to cope with waterlogging. To survive in such critical conditions, food crops must make complex anatomical, biochemical, and physiological adaptations. The plants' roots produce new adventitious roots, aerenchyma, and a barrier to radial oxygen loss as part of the morpho-

logical resistance mechanism. Tolerance to waterlogging is characterized by changes in a variety of physiological parameters, including photosynthesis, stomatal conductance, and gas exchange, as well as biochemical adaptations, including an increase in the quantity of fermentative enzymes, an energy crunch, and an increase in the supply of glycolysis.

Despite tremendous progress in understanding the subtleties of the waterlogging stress response, there is still a disconnect between laboratory-based studies and the stress tolerance of crops in the field. Only to a limited extent can phenotyping under controlled circumstances replicate what occurs in the field. To ensure a future with greater food security, it is crucial to bridge the gap between laboratory and field settings. In order to close this gap, advancements in genetic analysis in conjunction with plant phenotyping may be crucial.

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References

1. Angelopoulos, K.; Dichio, B.; Xiloyiannis, C. Inhibition of photosynthesis in olive trees (*Olea europaea* L.) during water stress and rewatering. *J. Exp. Bot.* **1996**, *47*, 1093–1100. [CrossRef]
2. Masson-Delmotte, V.; Zhai, P.; Pirani, A.; Connors, S.L.; Péan, C.; Berger, S.; Caud, N.; Chen, Y.; Goldfarb, L.; Gomis, M.I.; et al. (Eds.) IPCC 2021: Climate change 2021: The physical science basis. In *Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*; Cambridge University Press: Cambridge, UK, 2021.
3. IPCC. Climate change. The physical science basis. Working group I technical support unit. In *Working Group I Contribution to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*; Stocker, T.F., Qin, D., Plattner, G.-P., Tignor, M.M.B., Allen, A.K., Boschung, J., Nauels, A., Xia, Y., Bex, V., Midgley, P.M., Eds.; Cambridge University Press: Cambridge, UK, 2013.
4. Sharma, S.; Sharma, J.; Soni, V.; Kalaji, H.M.; Elsheery, N.I. Waterlogging tolerance: A review on regulative morpho- physiological homeostasis of crop plants. *J. Water Land Dev.* **2021**, *49*, 16–28. [CrossRef]
5. Westra, S.; Fowler, H.J.; Evans, J.P.; Alexander, L.V.; Berg, P.; Johnson, F.; Kendon, E.J.; Lenderink, G.; Roberts, N.M. Future changes to the intensity and frequency of short-duration extreme rainfall. *Rev. Geophys.* **2014**, *52*, 522–555. [CrossRef]
6. Price, R.K.; Vojinovic, Z. Urban flood disaster management. *Urban Water J.* **2008**, *5*, 259–276. [CrossRef]
7. Ziegler, V.H.; Ploschuk, E.; Weibel, A.; Insausti, P. Short-term responses to flooding stress of three Prunus rootstocks. *Sci. Hortic.* **2017**, *224*, 135–141. [CrossRef]
8. Kreuzwieser, J.; Rennenberg, H. Molecular and physiological responses of trees to waterlogging stress. *Plant Cell Environ.* **2014**, *37*, 2245–2259. [CrossRef]
9. Fischer, G.; Casierra-Posada, F.; Blanke, M. Impact of waterlogging on fruit crops in the era of climate change, with emphasis on tropical and subtropical species: A review. *Agron. Colomb.* **2023**, *41*, e108351. [CrossRef]
10. Insausti, P.; Gorjón, S. Floods affect physiological and growth variables of peach trees (*Prunus persica* (L.) Batsch), as well as the postharvest behavior of fruits. *Sci. Hortic.* **2013**, *152*, 56–60. [CrossRef]
11. Bhusal, N.; Seok Kim, H.; Han, S.G.; Yoon, T.M. Photosynthetic traits and plant- water relations of two apple cultivars grown as bi-leader tress under long- term waterlogging conditions. *Environ. Exp. Bot.* **2020**, *176*, 104–111. [CrossRef]
12. Ma, S.; Gai, P.; Geng, B.; Wang, Y.; Ullah, N.; Zhang, W.; Zhang, H.; Fan, Y.; Huang, Z. Exogenous melatonin improves waterlogging tolerance in wheat through promoting antioxidant enzymatic activity and carbon assimilation. *Agronomy* **2022**, *12*, 2876. [CrossRef]
13. Yeung, E.; van Veen, H.; Vashisht, D.; Sobral Paiva, A.L.S.; Hummel, M.; Rankenberg, T.; Steffens, B.; Steffen-Heins, A.; Sauter, M.; de Vries, M.; et al. A stress recovery signaling network for enhanced flooding tolerance in *Arabidopsis thaliana*. *Proc. Natl. Acad. Sci. USA* **2018**, *115*, E6085–E6094. [CrossRef] [PubMed]
14. Alvino, A.; Magliulo, V.; Zerbi, G. Problems of peach (*Prunus persica*) tolerance to anaerobic conditions due to excess soil water. *Riv. Ortoflorofruttic. Ital.* **1986**, *70*, 263–270.
15. Zhang, X.; Zhou, G.; Shabala, S.; Koutoulis, A.; Shabala, L.; Johnson, P.; Li, C.; Zhou, M. Identification of aerenchyma formation-related QTL in barley that can be effective in breeding for waterlogging tolerance. *Theor. Appl. Genet.* **2016**, *129*, 1167–1177. [CrossRef] [PubMed]
16. Bailey-Serres, J.; Lee, S.C.; Brinton, E. Water proofing crops: Effective flooding survival strategies. *Plant Physiol.* **2012**, *160*, 1698–1709. [CrossRef] [PubMed]
17. Ferronato, C.; Marinari, S.; Francioso, O.; Bello, D.; Trasarcepeda, C.; Antisari, L.V. Effect of waterlogging on soil biochemical properties and organic matter quality in different salt marsh systems. *Geoderma* **2019**, *338*, 302–312. [CrossRef]

18. Pezeshki, S.R.; DeLaune, R.D. Soil oxidation-reduction in wetlands and its impact on plant functioning. *Biology* **2012**, *1*, 196–221. [CrossRef] [PubMed]
19. Kaur, G.; Singh, G.; Motavalli, P.P.; Nelson, K.A.; Orlowski, J.M.; Golden, B.R. Impacts and management strategies for crop production in waterlogged or flooded soils: A review. *Agron. J.* **2020**, *112*, 1475–1501. [CrossRef]
20. Striker, G.G. Flooding Stress on Plants: Anatomical, Morphological and Physiological Responses. In *Botany*; Mworia, J., Ed.; InTech: Takasago, Japan, 2012; ISBN 978-953-51-0355-4.
21. Pucciariello, C.; Perata, P. Flooding tolerance in plants. In *Plant Stress Physiology*; Shabala, S., Ed.; CAB International: Wallingford, UK, 2012; pp. 148–170.
22. Larcher, W. *Physiological Plant Ecology*; Springer: Cham, Switzerland, 2003.
23. Hossain, A.; Uddin, S.N. Mechanisms of waterlogging tolerance in wheat: Morphological and metabolic adaptations under hypoxia or anoxia. *Aust. J. Crop Sci.* **2011**, *5*, 1094–1101.
24. Xie, L.-J.; Zhou, Y.; Chen, Q.-F.; Xiao, S. New insights into the role of lipids in plant hypoxia responses. *Prog. Lipid Res.* **2021**, *81*, 101072. [CrossRef]
25. Blom, C.W.P.M.; Voesenek, L.A.C.J. Flooding: The survival strategies of plants. *Trends Ecol. Evol.* **1996**, *11*, 290–295. [CrossRef]
26. Schopfer, P.; Brennicke, A. *Pflanzenphysiologie. 7. Auflage*; Spektrum Akademischer Verlag: Heidelberg, Germany, 2010.
27. Crane, J.H.; Balerdi, C.F.; Schaffer, B. *Managing Your Tropical Fruit Grove under Changing Water Table Levels*; Doc. HS957; Horticultural Sciences Department, UF/IFAS Extension: Gainesville, FL, USA, 2020.
28. Khondaker, N.A.; Ozawa, K. Papaya plant growth as affected by soil air oxygen deficiency. *Acta Hortic.* **2007**, *740*, 225–232. [CrossRef]
29. Kozlowski, T.; Pallardy, S. *Physiology of Woody Plants*, 2nd ed.; Academic Press: London, UK, 1997.
30. Lambers, H.; Oliveira, R.S. *Plant Physiological Ecology*; Springer Nature Switzerland AG: Cham, Switzerland, 2019.
31. Aldana, F.; García, P.N.; Fischer, G. Effect of waterlogging stress on the growth, development and symptomatology of cape gooseberry (*Physalis peruviana* L.) plants. *Rev. Acad. Colomb. Cienc. Exactas Fis. Nat.* **2014**, *38*, 393–400. [CrossRef]
32. Fischer, G.; Ramírez, F.; Casierra-Posada, F. Ecophysiological aspects of fruit crops in the era of climate change. A review. *Agron. Colomb.* **2016**, *34*, 190–199. [CrossRef]
33. Arguello, M.N.; Mason, R.E.; Roberts, T.L.; Subramanian, N.; Acuna, A.; Addison, C.K.; Lozada, D.N.; Miller, R.G.; Gbur, E. Performance of soft red winter wheat subjected to field soil waterlogging: Grain yield and yield components. *Field Crop. Res.* **2016**, *194*, 57–64. [CrossRef]
34. Herzog, M.; Striker, G.G.; Colmer, T.D.; Pedersen, O. Mechanisms of waterlogging tolerance in wheat—A review of root and shoot physiology. *Plant Cell Environ.* **2016**, *39*, 1068–1086. [CrossRef] [PubMed]
35. Zhang, Y.; Chen, Y.; Lu, H.; Kong, X.; Dai, J.; Li, Z.; Dong, H. Growth: Lint yield and changes in physiological attributes of cotton under temporal waterlogging. *Field Crop. Res.* **2016**, *194*, 83–93. [CrossRef]
36. Striker, G.G.; Colmer, T.D. Flooding tolerance of forage legumes. *J. Exp. Bot.* **2017**, *68*, 1851–1872. [CrossRef]
37. Wang, H.; Chen, Y.; Hu, W.; Wang, S.; Snider, J.L.; Zhou, Z. Carbohydrate metabolism in the subtending leaf cross-acclimates to waterlogging and elevated temperature stress and influences boll biomass in cotton (*Gossypium hirsutum*). *Physiol. Plant.* **2017**, *161*, 339–354. [CrossRef] [PubMed]
38. Schaffer, B.; Andersen, P.C.; Ploetz, R.C. Responses of fruit crops to flooding. *Hortic. Rev.* **1992**, *13*, 257–313.
39. Bailey-Serres, J.; Voesenek, L.A.C.J. Flooding stress: Acclimations and genetic diversity. *Annu. Rev. Plant Biol.* **2008**, *59*, 313–339. [CrossRef]
40. Sairam, R.K.; Dharmar, K.; Chinnusamy, V.; Meena, R.C. Waterlogging-induced increase in sugar mobilization, fermentation, and related gene expression in the roots of mung bean (*Vigna radiata*). *J. Plant Physiol.* **2008**, *166*, 602–616. [CrossRef] [PubMed]
41. Taiz, L.; Zeiger, E. *Plant Physiology*, 5th ed.; Sinauer Associates: Sunderland, MA, USA, 2010.
42. Jiménez, J.C.; Moreno, L.P.; Magnitskiy, S. Respuesta de las plantas a estrés por inundación. Una revisión. *Rev. Colomb. Cienc. Hortic.* **2012**, *6*, 96–109. [CrossRef]
43. Rasheeda, R.; Iqbala, M.; Ashrafa, M.A.; Hussaina, I.; Shafiqa, F.; Yousafa, A.; Zaheerb, A. Glycine betaine counteracts the inhibitory effects of waterlogging on growth, photosynthetic pigments, oxidative defense system, nutrient composition, and fruit quality in tomato. *J. Hortic. Sci. Biotechnol.* **2018**, *93*, 385–391. [CrossRef]
44. Schaffer, B.; Davies, F.; Crane, J.H. Responses to tropical and subtropical fruit trees to flooding in calcareous soil. *HortScience* **2006**, *41*, 549–555. [CrossRef]
45. Moreno, A.; Fischer, G. Efectos del anegamiento en los frutales. Una revisión. *Temas Agrar.* **2014**, *19*, 108–125. [CrossRef]
46. Blanke, M.M.; Cooke, D.T. Effects of flooding and drought on stomatal activity, transpiration, photosynthesis, water potential and water channel activity in strawberry stolons and leaves. *Plant Growth Regul.* **2004**, *42*, 153–160. [CrossRef]
47. Pallardy, S.G. *Physiology of Woody Plants*, 3rd ed.; Elsevier: Amsterdam, The Netherlands, 2008.
48. Else, M.A.; Hall, K.C.; Arnold, G.M.; Davies, W.J.; Jackson, M.B. Export of abscisic acid, 1-aminocyclopropane1-carboxylic acid, phosphate, and nitrate from roots to shoots of flooded tomato plants. *Plant Physiol.* **1995**, *107*, 377–384. [CrossRef] [PubMed]
49. Menezes-Silva, P.E.; Loram-Lourenço, L.; Alves, R.D.F.B.; Sousa, L.F.; Almeida, S.E.S.; Farnese, F.S. Different ways to die in a changing world: Consequences of climate change for tree species performance and survival through an ecophysiological perspective. *Ecol. Evol.* **2019**, *9*, 11979–11999. [CrossRef]

50. Parent, C.; Capelli, N.; Berger, A.; Crèvecoeur, M.; Dat, J.F. An overview of plant responses to soil waterlogging. *Plant Stress* **2008**, *2*, 20–27.
51. Sanclemente, M.A.; Schaffer, B.; Gil, P.M.; Vargas, A.I.; Davies, F.S. Pruning after flooding hastens recovery of flood-stressed avocado (*Persea americana* Mill.) trees. *Sci. Hortic.* **2014**, *169*, 27–35. [CrossRef]
52. Pezeshki, S. Wetland plant responses to soil flooding. *Environ. Exp. Bot.* **2001**, *46*, 299–312. [CrossRef]
53. Ashraf, M.A. Waterlogging stress in plants: A review. *Afr. J. Agric. Res.* **2012**, *7*, 1976–1981.
54. Shao, G.C.; Lan, J.J.; Yu, S.E.; Liu, N.; Guo, R.Q.; She, D.L. Photosynthesis and growth of winter wheat in response to waterlogging at different growth stages. *Photosynthetica* **2013**, *51*, 429–437. [CrossRef]
55. McGee, T.; Shahid, M.A.; Beckman, T.G.; Chaparro, J.X.; Scaffer, B.; Sarkhosh, A. Physiological and biochemical characterization of six Prunus rootstocks in response to flooding. *Environ. Exp. Bot.* **2021**, *183*, 104368. [CrossRef]
56. Yu, F.; Han, X.; Geng, C.; Zhao, Y.; Zhang, Z.; Qiu, F. Comparative proteomic analysis revealing the complex network associated with waterlogging stress in maize (*Zea mays* L.) seedling root cells. *Proteomics* **2015**, *15*, 135–147. [CrossRef] [PubMed]
57. Barickman, T.C.; Simpson, C.R.; Sams, C.E. Waterlogging causes early modification in the physiological performance, carotenoids, chlorophylls, proline, and soluble sugars of cucumber plants. *Plants* **2019**, *8*, 160. [CrossRef] [PubMed]
58. Marashi, S.K. Evaluation of uptake rate and distribution of nutrient ions in wheat (*Triticum aestivum* L.) under waterlogging condition. *Iran. J. Plant Physiol.* **2018**, *8*, 2539–2547.
59. Ploschuk, R.A.; Miralles, D.J.; Colmer, T.D.; Ploschuk, E.L.; Striker, G.G. Waterlogging of winter crops at early and late stages: Impacts on leaf physiology, growth, and yield. *Front. Plant Sci.* **2018**, *9*, 1863. [CrossRef] [PubMed]
60. Lin, H.H.; Lin, K.H.; Syu, J.Y.; Tang, S.Y.; Lo, H.F. Physiological and proteomic analysis in two wild tomato lines under waterlogging and high temperature stress. *J. Plant Biochem. Biotechnol.* **2016**, *25*, 87–96. [CrossRef]
61. Ren, B.; Zhu, Y.; Zhang, J.; Dong, S.; Liu, P.; Zhao, B. Effects of spraying exogenous hormone 6-benzyladenine (6-BA) after waterlogging on grain yield and growth of summer maize. *Field Crop. Res.* **2016**, *188*, 96–104. [CrossRef]
62. Zhang, F.; Zhu, K.; Wang, Y.Q.; Zhang, Z.P.; Lu, F.; Yu, H.Q.; Zou, J.Q. Changes in photosynthetic and chlorophyll fluorescence characteristics of sorghum under drought and waterlogging stress. *Photosynthetica* **2019**, *57*, 1156–1164. [CrossRef]
63. Lal, M.; Kumari, A.; Sheokand, S. Reactive oxygen species, reactive nitrogen species and oxidative metabolism under waterlogging stress. In *Reactive Oxygen, Nitrogen and Sulfur Species in Plants: Production, Metabolism, Signaling and Defense Mechanisms*; Hasanuzzaman, M., Fotopoulos, V., Nahar, K., Fujita, M., Eds.; John Wiley Sons Ltd.: Hoboken, NJ, USA, 2019; Chapter 34; pp. 777–812.
64. Liu, P.; Sun, F.; Gao, R.; Dong, H. RAP2. 6L overexpression delays waterlogging induced premature senescence by increasing stomatal closure more than antioxidant enzyme activity. *Plant Mol. Biol.* **2012**, *79*, 609–622. [CrossRef] [PubMed]
65. Voesenek, L.A.; Bailey-Serres, J. Flooding tolerance: O_2 sensing and survival strategies. *Curr. Opin. Plant Biol.* **2013**, *16*, 647–653. [CrossRef]
66. Tamang, B.G.; Magliozzi, J.O.; Maroof, M.S.; Fukao, T. Physiological and transcriptomic characterization of submergence and reoxygenation responses in soybean seedlings. *Plant Cell Environ.* **2014**, *37*, 2350–2365. [CrossRef] [PubMed]
67. Pan, J.; Sharif, R.; Xu, X.; Chen, X. Mechanisms of waterlogging tolerance in plants: Research progress and prospects. *Front. Plant Sci.* **2021**, *11*, 627331. [CrossRef] [PubMed]
68. Perez-Jimenez, M.; Perez-Tornero, O. Short term waterlogging in Citrus rootstocks. *Plants* **2021**, *10*, 2772. [CrossRef] [PubMed]
69. Childers, N.F.; White, D.G. Some physiological effects of excess soil moisture on Stayman Winesap apple trees. Ohio Agricultural Experiment Station Wooster, Ohio. *Res. Bull.* **1950**, *694*, 1–36.
70. Rodrigues, M.Â.; Raimundo, S.; Arrobas, M. Cover cropping in rainfed fruticulture. *World J. Agric. Soil Sci* **2019**, *1*, 1–3. [CrossRef]
71. Aragues, R.; Puy, J.; Isidoro, D. Vegetative growth response of young olive trees (*Olea europaea* L., cv. Arbequina) to soil salinity and waterlogging. *Plant Soil* **2004**, *258*, 69–80. [CrossRef]
72. International Olive Council. *Production Techniques in Olive Growing*; International Olive Council: Madrid, Spain, 2007; pp. 32–33.
73. Friedrich, G.; Fischer, M. *Physiologische Grundlagen des Obstbaues*; Verlag Eugen Ulmer: Stuttgart, Germany, 2000.
74. Paull, R.E.; Duarte, O. *Tropical fruits*, 2nd ed.; CAB International: Wallingford, UK, 2012; Volume 2.
75. Langan, P.; Bernad, V.; Walsh, J.; Henchy, J.; Khodaeiaminjan, M.; Mangina, E.; Nergao, S. Phenotyping for waterlogging tolerance in crops: Current trends and future prospects. *J. Exp. Bot.* **2022**, *73*, 5149–5169. [CrossRef]
76. Sathi, K.S.; Masud, A.A.C.; Anee, T.I.; Rahman, K.; Ahmed, N.; Hasanuzzaman, M. Soybean plants under waterlogging stress: Responses and adaptation mechanisms. In *Managing Plant Production under Changing Environment*; Hasanuzzaman, M., Ahammed, G.J., Nahar, K., Eds.; Springer Nature: Cham, Switzerland, 2022; pp. 103–134.
77. Panozzo, A.; Dal Cortivo, C.; Ferrari, M.; Vicelli, B.; Varotto, S.; Vamerali, T. Morphological changes and expressions of AOX1A, CYP81D8, and putative PFP genes in a large set of commercial maize hybrids under extreme waterlogging. *Front. Plant Sci.* **2019**, *10*, 62. [CrossRef] [PubMed]
78. Anee, T.I.; Nahar, K.; Rahman, A.; Mahmud, J.A.; Bhuiyan, T.F.; Alam, M.U.; Fujita, M.; Hasanuzzaman, M. Oxidative damage and antioxidant defense in *Sesamum indicum* after different waterlogging durations. *Plants* **2019**, *8*, 196. [CrossRef]
79. Sauter, M. Root responses to flooding. *Curr. Opin. Plant Biol.* **2013**, *16*, 282–286. [CrossRef] [PubMed]
80. Voesenek, L.A.C.J.; Sasidharan, R. Ethylene—And oxygen signaling—Drive plant survival during flooding. *Plant Biol.* **2013**, *15*, 426–435. [CrossRef] [PubMed]
81. Loreti, E.; van Veen, H.; Perata, P. Plant responses to flooding stress. *Curr. Opin. Plant Biol.* **2016**, *33*, 64–71. [CrossRef] [PubMed]

82. Joo, J.H.; Bae, Y.S.; Lee, J.S. Role of auxin-induced reactive oxygen species in root gravitropism. *Plant Physiol.* **2001**, *126*, 1055–1060. [CrossRef] [PubMed]
83. Sagi, M.; Fluhr, R. Production of reactive oxygen species by plant NADPH oxidases. *Plant Physiol.* **2006**, *141*, 336–340. [CrossRef]
84. Hoidal, J.R. Reactive oxygen species and cell signaling. *Am. J. Respir. Cell Mol. Biol.* **2001**, *25*, 661–663. [CrossRef] [PubMed]
85. Mhamdi, A.; van Breusegem, F. Reactive oxygen species in plant development. *Development* **2018**, *145*, dev164376. [CrossRef]
86. Mittler, R.; Vanderauwera, S.; Gollery, M.; Breusegem, F. Reactive oxygen gene network of plants. *Trends Plant Sci.* **2004**, *9*, 490–498. [CrossRef]
87. Baxter-Burrell, A.; Yang, Z.; Springer, P.S.; Bailey-Serres, J. RopGAP4 dependent Rop GTPase rheostat control of *Arabidopsis* oxygen deprivation tolerance. *Science* **2002**, *296*, 2026–2028. [CrossRef] [PubMed]
88. Neill, S.J.; Desikan, R.; Clarke, A.; Hurst, R.D.; Hancock, J.T. Hydrogen peroxide and nitric oxide as signaling molecules in plants. *J. Exp. Bot.* **2002**, *53*, 1237–1247. [CrossRef] [PubMed]
89. Christianson, J.A.; Llewellyn, D.J.; Dennis, E.S.; Wilson, I.W. Global gene expression responses to waterlogging in roots and leaves of cotton (*Gossypium hirsutum* L.). *Plant Cell Physiol.* **2010**, *51*, 21–37. [CrossRef] [PubMed]
90. Teoh, E.Y.; Teo, C.H.; Baharum, N.A.; Pua, T.-L.; Tan, B.C. Waterlogging stress induces antioxidant defense responses, aerenchyma formation and alters metabolisms of banana plants. *Plants* **2022**, *11*, 2052. [CrossRef]
91. Jiang, M.; Zhang, J. Water stress-induced abscisic acid accumulation triggers the increased generation of reactive oxygen species and up-regulates the activities of antioxidant enzymes in maize leaves. *J. Exp. Bot.* **2002**, *53*, 2401–2410. [CrossRef] [PubMed]
92. Dawood, T.; Yang, X.; Visser Eric, J.W.; Beek, A.H.; Kensche, P.R.; Cristescu, S.; Lee, S.; Flokova, K.; Nguyen, D.; Mariani, C.; et al. A co-opted hormonal cascade activates dormant adventitious root primordia upon flooding in *Solanum dulcamara*. *Plant Physiol.* **2016**, *170*, 2351–2364. [CrossRef]
93. Nguyen, T.-N.; Son, S.; Jordan, M.C.; Levin, D.B.; Ayele, B.T. Lignin biosynthesis in wheat (*Triticum aestivum* L.): Its response to waterlogging and association with hormonal levels. *BMC Plant Biol.* **2016**, *16*, 28. [CrossRef] [PubMed]
94. Zhang, P.; Lyu, D.; Jia, L.; He, J.; Qin, S. Physiological and de novo transcriptome analysis of the fermentation mechanism of *Cerasus sachalinensis* roots in response to short-term waterlogging. *BMC Genom.* **2017**, *18*, 649. [CrossRef] [PubMed]
95. Zeng, W.; Mostafa, S.; Lu, Z.; Jin, B. Melatonin-Mediated Abiotic Stress Tolerance in Plants. *Front. Plant Sci.* **2022**, *13*, 847175. [CrossRef]
96. Moustafa-Farag, M.; Mahmoud, A.; Arnao, M.B.; Sheteiwy, M.S.; Dafea, M.; Soltan, M.; Elkelish, A.; Hasanuzzaman, M.; Ai, S. Melatonin-induced water stress tolerance in plants: Recent advances. *Antioxidants* **2020**, *9*, 809. [CrossRef]
97. Gu, X.; Xue, L.; Lu, L.; Xiao, J.; Song, G.; Xie, M.; Zhang, H. Melatonin enhances the waterlogging tolerance of *Prunus persica* by modulating antioxidant metabolism and anaerobic respiration. *J. Plant Growth Regul.* **2021**, *40*, 2178–2190. [CrossRef]
98. Rodriguez, M.E.; Doffo, G.N.; Cerrillo, T.; Luquez, V.M. Acclimation of cuttings from different willow genotypes to flooding depth level. *New For.* **2018**, *49*, 415–427. [CrossRef]
99. Cooke, J.E.K.; Martin, T.A.; Davis, J.M. Short term physiological and developmental response to nitrogen availability in hybrid poplar. *New Phytol.* **2005**, *167*, 41. [CrossRef]
100. Armstrong, W.; Beckett, P.M.; Colmer, T.D.; Setter, T.L.; Greenway, H. Tolerance of roots to low oxygen: 'Anoxic' cores, the phytoglobin-nitric oxide cycle, and energy or oxygen sensing. *J. Plant Physiol.* **2019**, *239*, 92–108. [CrossRef]
101. Schwarz, D.; Rouphael, Y.; Colla, G.; Venema, J. Grafting as a tool to improve tolerance of vegetables to abiotic stresses: Thermal stress, water stress and organic pollutants. *Sci. Hortic.* **2010**, *127*, 162–171. [CrossRef]
102. Orazem, P.; Stampar, F.; Hudina, M. Quality analysis of 'Redhaven' peach fruit grafted on 11 rootstocks of different genetic origin in a replant soil. *Food Chem.* **2011**, *124*, 1691–1698. [CrossRef]
103. Reig, G.; Zarrouk, O.; Font i Forcada, C.; Moreno, M.Á. Anatomical graft compatibility study between apricot cultivars and different plum based rootstocks. *Sci. Hortic.* **2018**, *237*, 67–73. [CrossRef]
104. Wistrom, B.; Emilsson, T.; Sjiman, H.; Levinsson, A. Experimental evaluation of waterlogging and drought tolerance of essential Prunus species in central Europe. *For. Ecol. Manag.* **2023**, *537*, 120904. [CrossRef]
105. Kongsri, S.; Nartvaranant, P.; Boonprakob, U. A comparison of flooding tolerance of guava tree propagated from shoot layering and seedling. *Acta Hortic.* **2020**, *1298*, 625–632. [CrossRef]
106. Thani, Q.A.; Vargas, A.I.; Schaffer, B.; Liu, G.; Crane, J.D. Responses of papaya plants in a potting medium in containers to flooding and solid oxygen fertilization. In Proceedings of the Florida State Horticultural Society; Florida State Horticultural Society: Orlando, FL, USA, 2016; Volume 129, pp. 27–34.
107. Liu, G.; Porterfield, D.M. Oxygen enrichment with magnesium peroxide for minimizing hypoxic stress of flooded corn. *J. Plant Nutr. Soil Sci.* **2014**, *177*, 733–740. [CrossRef]
108. Tuheteru, F.D.; Wu, Q.-S. Arbuscular mycorrhizal fungi and tolerance of waterlogging stress in plants. In *Arbuscular Mycorrhizas and Stress Tolerance of Plants*; Wu, Q.-S., Ed.; Springer: Singapore, 2017; pp. 43–66.
109. Chebet, D.; Kariuki, W.; Wamocho, L.; Rimberia, F. Effect of arbuscular mycorrhizal inoculation on growth, biochemical characteristics and nutrient uptake of passion fruit seedlings under flooding stress. *Int. J. Agron. Agric. Res.* **2020**, *16*, 24–31.
110. Castro-Duque, N.E.; Chávez-Arias, C.C.; Restrepo-Díaz, H. Foliar glycine betaine or hydrogen peroxide sprays ameliorate waterlogging stress in cape gooseberry. *Plants* **2020**, *9*, 644. [CrossRef] [PubMed]
111. Flórez-Velasco, N.; Balaguera-López, H.E.; Restrepo-Díaz, H. Effects of foliar urea application on lulo (*Solanum quitoense* cv. septentrionale) plants grown under different waterlogging and nitrogen conditions. *Sci. Hortic.* **2015**, *186*, 154–162. [CrossRef]

112. Sanchez-Reinoso, A.D.; Jimenez-Pulido, Y.; Martinez-Perez, J.P.; Pinilla, C.S.; Fischer, G. Chlorophyll fluorescence and other physiological parameters as indicators of waterlogging and shadow stress in lulo (*Solanum quitoense* var. septetrionale) seedlings. *Rev. Colomb. Cienc. Hortícolas* **2019**, *13*, 325–335. [CrossRef]
113. León-Burgos, A.F.; Unigarro, C.A.; Balaguera-López, H.E. Soil waterlogging conditions affect growth, water status, and chlorophyll "a" fluorescence in coffee plants (*Coffea arabica* L.). *Agronomy* **2022**, *12*, 1270. [CrossRef]
114. Collaku, A.; Harrison, S. Losses in wheat due to waterlogging. *Crop Sci.* **2002**, *42*, 444–450. [CrossRef]
115. Setter, T.L.; Waters, I.; Sharma, S.K.; Singh, K.N.; Kulshreshtha, N.; Yaduvanshi, N.P.; Ram, P.C.; Singh, B.N.; Rane, J.; McDonald, G.; et al. Review of wheat improvement for waterlogging tolerance in Australia and India: The importance of anaerobiosis and element toxicities associated with different soils. *Ann. Bot.* **2009**, *103*, 221–235. [CrossRef]

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